

Climate 5000 Series - Gen 4

9K BTU Single Zone Regular System

4-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

9K Btu Single Zone Regular System - 4-Way Cassette IDU

Indoor part number 8-733-962-716

Indoor model number BMS500-AAU009-1AHCXD

Outdoor part number 8-733-962-684

Outdoor model number BMS500-AAS009-1CSXRD

Efficiency Data AHRI 210/240 2023

SEER2 21.6

EER2 14.3

HSPF2 Region IV 11.3

HSPF2 Region V 8.8

COP(47F) 4.00

COP (5F) 2.17

Performance Data AHRI 210/240 2023

Cooling	Capacity Range	3700~12000
	Rated Capacity	9000
	Current	4.5
	Input Power	629

Heating	Capacity Range	4700~13700
	Rated Capacity (47F)	10000
	Rated Capacity (5F)	8300
	Current	3.3
	Input Power	733

Electrical Data

Power supply (V / Ø / Hz) 208-230 / 1 / 60

Min. circuit ampacity indoor/outdoor (A) 3.0 / 12.0

Max. fuse indoor/outdoor (A) 15.0 / 15.0

Compressor Data

Type: Rotary

Refrigerant oil /oil charge (ml) VG74 / 310

Input (W) 325 / 826

Rated current (RLA/A) 2.40 / 5.65

Piping

Refrigerant precharge (oz/kg) 32.45 / 0.92

Length for precharge (ft/m) 24.6 / 7.5

Liquid side (in/mm) 1/4" / Ø6.35

Gas side (in/mm) 3/8" / Ø9.52

Additional Charge of Refrigerant (oz/ft)/(g/m) 0.16 / 15

Max. refrigerant pipe length (ft/m) 82.00 / 25

Max. difference in height (ft/m) 49.20 / 15

Operation Range

ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type R454B

Design pressure (PSIG) 550 / 340

Dehumidification (pts/hr) 1.56

IDU	Fan motor model	ZKFN-45-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	600 / 530 / 380
	Air flow, Turbo / H / M / L / Si (CFM)	353.2 / 312.0 / 264.9 / 182.5 / 147.2
	Approx. Air Throw Range ft (m)	9.8 / 3
	Noise level, Turbo / H / M / L / Si (dB(A))	42.5 / 40 / 37.0 / 26.0 / 24.0
	Net weight (lbs / kg)	35.27 / 16.0
	Gross weight (lbs / kg)	41.89 / 19.0
	Panel net weight (lbs / kg)	5.95 / 2.7
ODU	Gross panel weight (lbs / kg)	9.48 / 4.3
	Drain pipe diameter (mm)	ODØ25mm
	Fan motor model	ZKFN-34-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed (RPM)	800 / 700 / 600
	Air flow, Max (CFM)	1236.1
	Noise level, Max (dB(A))	54.5
	Net weight (lbs / kg)	62.17 / 28.2
	Gross weight (lbs / kg)	67.68 / 30.7

Accessories

BMS-CCG-XXD (8733962708) Decorative Grille - 9K ~ 18K

BMS-WT2-XXD (8733962707) Wired Wall Thermostat

BMS-DRY-XXD (8733969263) 24V Interface

BMS-PK1-7X1 (8733951010) 25ft Lineset

Standard Parts

8-733-969-065 Remote Control

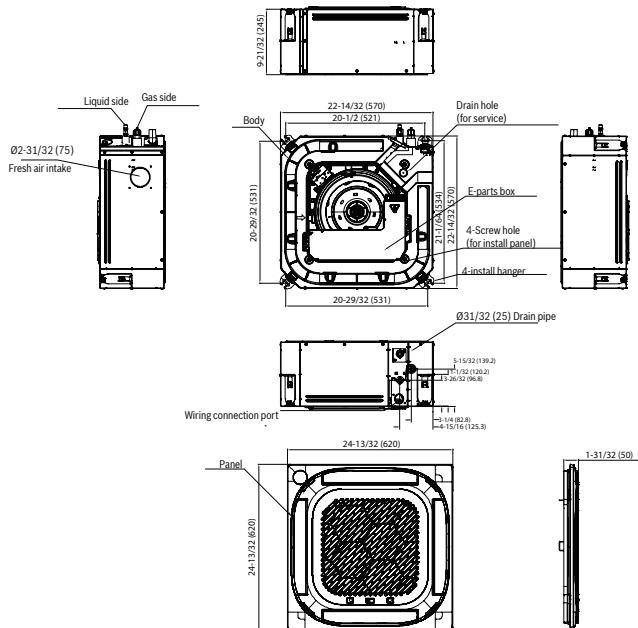
Climate 5000 Series - Gen 4

9K Btu Single Zone Regular System - 4-Way Cassette IDU



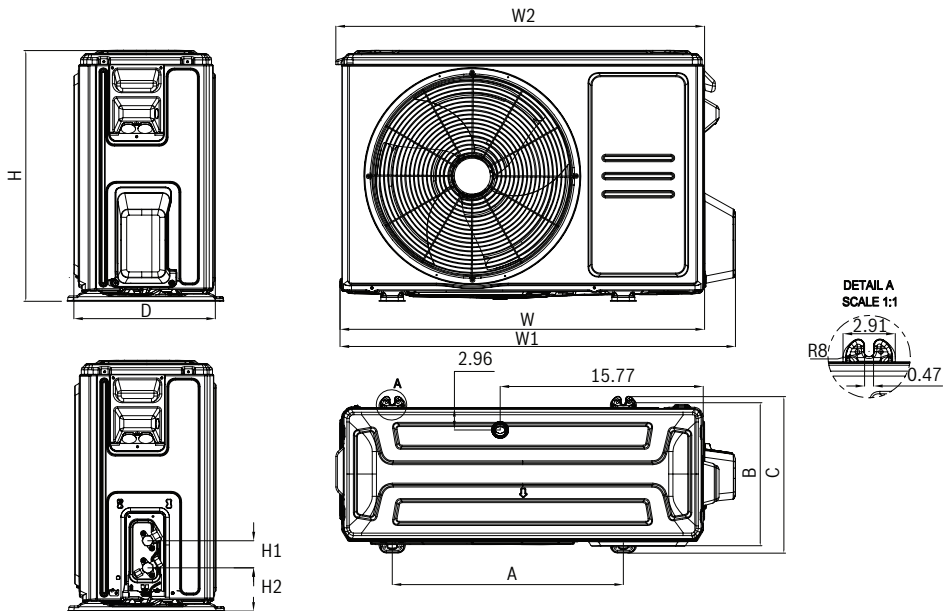
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Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU009-1AHCXD	22.44	22.44	9.65

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS009-1CSXRD	30.12	11.9	21.85	32.9	30.87	17.8	11.26	12.36	2.36	3.67

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Cooling Performance - Indoor Dry Bulb Temperature at 353 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	S/T	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
-13°F (-25.0°C)	TC	9772	9785	9796	9796	9810	10361	10373	10384	10386	10398	10980	10994	11007	11009	11021	11642	11654	11667	11669	11683
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.63	0.64	0.64	0.64	0.64	0.67	0.67	0.67	0.67	0.68	0.71	0.71	0.71	0.71	0.72	0.76	0.76	0.76	0.76	0.76
-4°F (-20.0°C)	TC	10524	10538	10549	10550	10564	11158	11171	11182	11185	11198	11825	11840	11853	11856	11869	12537	12551	12565	12566	12582
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	0.66	0.66	0.66	0.66	0.66	0.70	0.70	0.70	0.70	0.70	0.74	0.74	0.74	0.74	0.74	0.78	0.78	0.78	0.78	0.79
0°F (-17.8°C)	TC	10858	10873	10884	10885	10899	11512	11526	11537	11540	11553	12200	12216	12230	12232	12246	12935	12949	12963	12965	12982
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.67	0.67	0.67	0.67	0.67	0.71	0.71	0.71	0.71	0.71	0.75	0.75	0.75	0.75	0.75	0.80	0.80	0.80	0.80	0.80
5°F (-15.0°C)	TC	11276	11291	11303	11303	11319	11955	11969	11981	11983	11998	12669	12686	12700	12702	12717	13433	13447	13462	13464	13481
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	0.68	0.68	0.68	0.68	0.68	0.72	0.72	0.72	0.72	0.72	0.76	0.77	0.77	0.77	0.77	0.81	0.81	0.81	0.81	0.81
14°F (-10.0°C)	TC	10916	10931	10943	10943	10958	11574	11588	11599	11602	11615	12266	12282	12295	12298	12311	13005	13019	13033	13035	13051
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.63	0.63	0.63	0.63	0.63	0.67	0.67	0.67	0.67	0.67	0.71	0.71	0.71	0.71	0.71	0.75	0.75	0.75	0.75	0.75
17°F (-8.3°C)	TC	10797	10811	10823	10823	10838	11447	11461	11472	11474	11488	12131	12147	12160	12163	12176	12862	12876	12890	12892	12908
	S/T	0.62	0.72	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.88	0.55	0.64	0.73	0.73	0.83	0.52	0.61	0.69	0.69	0.79
	PI	0.61	0.62	0.62	0.62	0.62	0.65	0.65	0.65	0.65	0.65	0.69	0.69	0.69	0.69	0.69	0.73	0.73	0.73	0.73	0.73
23°F (-5.0°C)	TC	10981	10996	11008	11008	11023	11643	11657	11668	11671	11685	12339	12355	12369	12371	12385	13082	13096	13111	13113	13129
	S/T	0.61	0.72	0.81	0.81	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.64	0.65	0.65	0.65	0.65	0.68	0.68	0.68	0.69	0.69	0.72	0.73	0.73	0.73	0.73	0.77	0.77	0.77	0.77	0.77
35°F (1.7°C)	TC	11351	11366	11378	11379	11394	12035	12049	12061	12063	12078	12754	12771	12785	12787	12801	13522	13537	13552	13554	13571
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.70	0.71	0.71	0.71	0.71	0.75	0.75	0.75	0.75	0.75	0.79	0.79	0.79	0.79	0.79	0.84	0.84	0.84	0.84	0.84
41°F (5.0°C)	TC	10584	10599	10610	10611	10625	11222	11236	11247	11249	11262	11893	11908	11922	11924	11937	12609	12623	12637	12639	12655
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.69	0.69	0.78
	PI	0.55	0.55	0.55	0.55	0.55	0.59	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.62	0.62	0.66	0.66	0.66	0.66	0.66
47°F (8.3°C)	TC	9818	9831	9842	9842	9856	10410	10422	10432	10434	10447	11032	11046	11058	11060	11073	11696	11709	11722	11724	11738
	S/T	0.63	0.73	0.83	0.83	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.40	0.40	0.40	0.40	0.40	0.42	0.43	0.43	0.43	0.43	0.45	0.45	0.45	0.45	0.45	0.48	0.48	0.48	0.48	0.48
50°F (10.0°C)	TC	9775	9789	9799	9800	9813	10365	10377	10387	10389	10402	10984	10998	11010	11012	11025	11646	11658	11671	11673	11687
	S/T	0.62	0.73	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.89	0.55	0.65	0.73	0.73	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.41	0.41	0.41	0.41	0.41	0.44	0.44	0.44	0.44	0.44	0.46	0.46	0.46	0.46	0.46	0.49	0.49	0.49	0.49	0.49
59°F (15.0°C)	TC	9648	9661	9671	9672	9685	10229	10241	10251	10253	10266	10840	10854	10867	10869	10881	11494	11506	11519	11520	11535
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.44	0.44	0.44	0.44	0.44	0.47	0.47	0.47	0.47	0.47	0.50	0.50	0.50	0.50	0.50	0.53	0.53	0.53	0.53	0.53
68°F (20.0°C)	TC	9520	9533	9543	9544	9556	10094	10106	10116	10118	10130	10697	10711	10723	10725	10737	11341	11354	11366	11368	11382
	S/T	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78	0.48	0.57	0.65	0.65	0.74
	PI	0.47	0.47	0.47	0.47	0.48	0.50	0.50	0.50	0.50	0.50	0.53	0.53	0.53	0.53	0.53	0.56	0.56	0.57	0.57	0.57
77°F (25.0°C)	TC	9392	9405	9415	9416	9428	9958	9970	9980	9982	9994	10553	10567	10579	10581	10593	11189	11201	11214	11215	11229
	S/T	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.80	0.49	0.58	0.66	0.66	0.75	0.47	0.55	0.62	0.62	0.71
	PI	0.50	0.51	0.51	0.51	0.51	0.53	0.54	0.54	0.54	0.54	0.57	0.57	0.57	0.57	0.57	0.60	0.60	0.60	0.60	0.60
82°F (27.8°C)	TC	9518	9531	9542	9542	9555	10092	10104	10114	10116	10128	10695	10709	10721	10723	10735	11339	11352	11364	11366	11380
	S/T	0.56	0.66	0.75	0.75	0.85	0.53	0.62	0.70	0.70	0.80	0.50	0.58	0.66	0.66	0.76	0.47	0.55	0.63	0.63	0.71
	PI	0.56	0.57	0.57	0.57	0.57	0.60	0.60	0.60	0.60	0.60	0.63	0.64	0.64	0.64	0.64	0.67	0.67	0.67	0.67	0.67
86°F (30.0°C)	TC	9868	9881	9892	9892	9906	10463	10475	10485	10487	10500	11088	11102	11114	11117	11129	11756	11769	11781	11783	11798
	S/T	0.58	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79	0.49	0.58	0.65	0.65	0.75
	PI	0.67	0.67	0.67	0.67	0.67	0.71	0.71	0.71	0.71	0.71	0.75	0.75	0.75	0.75	0.75	0.79	0.80	0.80	0.80	0.80
95°F (35.0°C)	TC	10654	10668	10680	10680	10695	11296	11310	11321	11323	11336	11971	11987	12000	12002	12016	12692	12706	12720	12722	12738
	S/T	0.64	0.75	0.85	0.85	0.97	0.60	0.71	0.81	0.81	0.92	0.57	0.67	0.76	0.76	0.87	0.54	0.63	0.72	0.72	

Climate 5000 Series - Gen 4

9K Btu Single Zone Regular System - 4-Way Cassette IDU



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Heating Performance - Indoor Dry Bulb Temperature at 353 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	5025	4925	4900	4880	4836	0.93	0.94	0.94	0.94	0.94
-4°F(-20.0°C)	6256	6131	6100	6076	6021	0.98	0.99	0.99	1.00	1.00
0°F(-17.8°C)	7259	7113	7078	7049	6986	1.04	1.05	1.05	1.05	1.06
5°F(-15.0°C)	8512	8342	8300	8267	8192	1.11	1.12	1.12	1.12	1.13
14°F(-10.0°C)	9281	9095	9050	9014	8933	1.07	1.08	1.09	1.09	1.09
17°F(-8.3°C)	9537	9347	9300	9263	9179	1.06	1.07	1.07	1.08	1.08
22°F(-5.6°C)	10096	9894	9844	9805	9717	1.09	1.10	1.10	1.10	1.11
27°F(-2.8°C)	10654	10441	10389	10347	10254	1.11	1.12	1.12	1.13	1.13
32°F(-0.0°C)	11212	10988	10933	10889	10791	1.14	1.15	1.15	1.15	1.16
35°F(1.7°C)	11547	11316	11260	11215	11114	1.15	1.16	1.17	1.17	1.17
37°F(2.8°C)	11964	11725	11666	11620	11515	1.15	1.15	1.16	1.16	1.16
42°F(5.6°C)	13007	12747	12683	12632	12519	1.12	1.13	1.13	1.14	1.14
44.6°F(7.0°C)	13424	13156	13090	13038	12920	1.11	1.12	1.12	1.13	1.13
47°F(8.3°C)	14050	13769	13700	13645	13522	1.10	1.11	1.11	1.11	1.12
52°F(11.1°C)	13908	13630	13562	13508	13386	1.01	1.02	1.02	1.02	1.03
57°F(13.9°C)	13767	13492	13424	13370	13250	0.92	0.93	0.93	0.93	0.94
59°F(15.0°C)	13710	13436	13369	13315	13196	0.88	0.89	0.89	0.90	0.90
62°F(16.7°C)	13625	13353	13286	13233	13114	0.83	0.84	0.84	0.84	0.84
64.4°F(18.0°C)	13558	13286	13220	13167	13049	0.79	0.79	0.80	0.80	0.80
69.8°F(21.0°C)	13405	13137	13071	13019	12902	0.69	0.70	0.70	0.70	0.70
75.2°F(24.0°C)	13252	12987	12922	12870	12755	0.60	0.60	0.60	0.60	0.60

Bolded values were tested. All other values are simulated and for reference only.